Your dream, our reality.

An integrated, AI-powered smart home System focused on security, comfort, energy efficiency, and automation

FINAL PROJECT REPORT

Project Period: 9 November 2026 – 17 January 2027 (10 Weeks)

Team Member	Role
Eng. Mazen Adel	Project Manager
Eng. Mohamed Ayman	Software & Testing
Eng. Rahma Mohammed	Business Development
Eng. Rana Saber	Hardware & System Architecture
Eng. Gehad Mohammed	Security Engineer
Eng. Asmaa Khaled	Network & IoT

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1. Executive Summary

IHOME is an AI-powered smart home system designed to improve home security, comfort, energy efficiency, and automation. Its smart technologies are embedded directly into the home's infrastructure during construction, so that lighting, climate, security, and voice control work together as a single system managed from one mobile application.

This report documents the project from end to end: its mission and objectives, scope and deliverables, team and responsibilities, schedule, resources, budget, risk management, quality assurance, and communication plan. It closes with the testing and verification approach, a conclusion, and recommendations for future work.

1.1 Project at a Glance

Item	Details
Project Name	IHOME – AI-Powered Smart Home System
Project Duration	10 weeks (9 November 2026 – 17 January 2027)
Project Manager	Mazen
Team Size	6 members
Core Deliverables	Lighting System, Voice Control, Security System, Climate Control, Mobile Application, System Installation, Testing & Final Documentation
Service Packages	Essential, Plus, Premium

2. Project Overview

2.1 Mission

To build an intelligent, secure, and energy-efficient smart home by integrating advanced automation and smart technologies directly into the home's infrastructure during construction, ensuring comfort, safety, and seamless control.

2.2 Vision

To become a model for future-ready smart living by creating a fully connected, sustainable, and expandable home that enhances everyday life through intelligent technology, automation, and peace of mind.

2.3 Purpose & Strategic Alignment

To design and develop a fully integrated smart home where smart technologies are embedded into the home's infrastructure from the initial planning and construction stages, providing an intelligent, secure, energy-efficient, and comfortable living environment.

2.4 Core Values

Innovation	**Safety & Security**
Sustainability	**Comfort & Convenience**
Reliability	**Integration**
Scalability	**Privacy & Cybersecurity**
Peace of Mind	**Quality & Excellence**

3. Project Objectives

The project was set up to achieve the following measurable objectives, each supported by one or more of the project deliverables.

#	Objective	Delivered Through
1	Integrate 100% of the planned smart systems into the home's infrastructure during the construction phase.	System Installation
2	Reduce energy consumption by at least 30% through intelligent automation and monitoring.	Climate Control, Lighting System
3	Automate at least 5 major home functions, including lighting, temperature, curtains, security, and appliances.	Lighting System, Climate Control, Security System, Voice Control
4	Enable centralized control of all integrated systems through the IHOME app.	Mobile Application
5	Achieve a system response time of less than 2 seconds for standard control commands.	Mobile Application, Voice Control
6	Provide 24/7 security monitoring through smart sensors and detection systems.	Security System
7	Design an expandable infrastructure that supports at least 5 additional smart devices without major structural modifications.	System Installation, Network & IoT design

4. Scope & Deliverables

The project delivers seven main components, from the individual smart systems through installation, testing, and handover.

Deliverable	Description	Owner	Deadline
Lighting System	An automated lighting system that allows users to control lights remotely, schedule operation, and adjust lighting based on occupancy and natural light.	Rana	6 Dec 2026

Deliverable	Description	Owner	Deadline
Voice Control	A voice-activated system that allows users to control home functions and smart devices through voice commands.	Asmaa	3 Jan 2027
Security System	An integrated security system including smart cameras, motion detectors, smart locks, and real-time alerts to protect residents and monitor children and pets.	Gehad	13 Dec 2026
Climate Control	An automated temperature control system that monitors indoor conditions and adjusts air conditioning and heating to improve comfort and reduce energy consumption.	Rana	13 Dec 2026
Mobile Application	A centralized mobile application that enables users to monitor, control, and manage all connected smart home systems remotely through one platform.	Mohamed	27 Dec 2026
System Installation	Complete installation and integration of smart infrastructure, electrical wiring, networking equipment, sensors, controllers, and automation devices during the construction phase.	Rana	10 Jan 2027
Testing & Final Documentation	Comprehensive system testing to verify functionality, security, connectivity, and performance, followed by technical documentation and a final project report.	Mohamed (testing) Rahma (documentation)	17 Jan 2027

5. Stakeholders

Internal Stakeholders	External Stakeholders
Project Manager Engineers Developers Construction Team	Project Sponsor Homeowners Suppliers Technology Vendors

6. Team & Responsibilities

The project was delivered by a team of six members, each responsible for a defined area of the system.

Team Member	Role	Main Responsibilities
Mazen	Project Manager	Planning, coordination, and monitoring progress
Mohamed	Software & Testing	Mobile application, testing, and bug fixing
Rahma	Business Development	Suppliers, procurement, and documentation
Rana	Hardware & System Architecture	Smart lighting, climate control, and installation
Gehad	Security Engineer	Smart security, risk assessment, and cybersecurity
Asmaa	Network & IoT	Network design, device connectivity, and voice control

7. Project Schedule

The project ran for 10 weeks, from 9 November 2026 to 17 January 2027. Many tasks were planned to run in parallel: architecture, security review, procurement, and network planning overlap, and app development overlaps with the lighting, security, climate, and connectivity work.

7.1 Task Schedule

Task	Assigned To	Duration	Deadline
Project requirements & stakeholder approval	Mazen	7 days	15 Nov 2026
System architecture & device selection	Rana	10 days	22 Nov 2026
Security requirements & risk review	Gehad	7 days	15 Nov 2026
Supplier research & procurement	Rahma	10 days	29 Nov 2026
Network & IoT design	Asmaa	10 days	29 Nov 2026
Smart lighting system	Rana	14 days	6 Dec 2026
Smart security system	Gehad	21 days	13 Dec 2026
Climate control system	Rana	14 days	13 Dec 2026
Mobile application development	Mohamed	28 days	27 Dec 2026
Network & IoT device connectivity	Asmaa	21 days	27 Dec 2026
Voice control integration	Asmaa	14 days	3 Jan 2027
System installation & full integration	Rana	14 days	10 Jan 2027
Cybersecurity configuration & review	Gehad	14 days	10 Jan 2027
System testing & defect fixing	Mohamed	14 days	17 Jan 2027
Final documentation & stakeholder handover	Rahma	7 days	17 Jan 2027

7.2 Project Timeline (10-Week Gantt Chart)

W1 starts 9 Nov 2026 and W10 starts 11 Jan 2027. Blue cells show the weeks in which each task is active.

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Project requirements & stakeholder approval	■									
System architecture & device selection	■	■								
Security requirements & risk review	■									
Supplier research & procurement		■	■							
Network & IoT design		■	■							
Smart lighting system			■	■						
Smart security system			■	■	■					
Climate control system				■	■					
Mobile application development				■	■	■	■			
Network & IoT device connectivity					■	■	■			
Voice control integration							■	■		
System installation & full integration								■	■	
Cybersecurity configuration & review								■	■	
System testing & defect fixing									■	■
Final documentation & handover										■

8. Resource Allocation

8.1 Project Resources

Resource Category	Available Resources
Human Resources	6 team members: Mazen, Mohamed, Rahma, Rana, Gehad, Asmaa
Material Resources	Smart sensors, cameras, smart locks, smart lights, temperature sensors, cables
Equipment Resources	Laptops, IoT devices, Wi-Fi router, testing equipment, installation tools
Financial Resources	Budget for hardware, software, procurement, installation, and internet
Time	10 weeks (9 November 2026 – 17 January 2027)

8.2 Allocation by Task

Task	Responsible	Materials / Equipment	Cost Category
Project planning & requirements	Mazen	Laptop, documentation tools	—
Hardware & system architecture design	Rana	Laptop, sensors, IoT devices	—
Network design & IoT connectivity	Asmaa	Wi-Fi router, network tools	Internet costs
Smart lighting development	Rana	Smart lights, controllers, sensors	Hardware costs
Smart security system development	Gehad	Cameras, smart locks, motion sensors	Hardware costs
Climate control implementation	Rana	Temperature sensors, smart thermostat	Hardware costs
Mobile application development	Mohamed	Laptop, development tools, database	Software costs (if any)
Voice control integration	Asmaa	Microphone, IoT devices, AI tools	Integration costs
Cybersecurity assessment	Gehad	Security testing tools, laptop	—
System integration & testing	Mohamed, Rana, Asmaa	All smart devices, laptops, testing tools	—
Final documentation & project review	Rahma, Mazen	Laptop, documentation tools	—
Final deployment & presentation	All team members	Complete smart home system	Installation costs

8.3 Resource Allocation Strategy

- Assign tasks according to each member's skills and responsibilities.
- Allow parallel work whenever possible.
- Ensure hardware and software are available before starting tasks.
- Coordinate between the software, hardware, security, and network areas.
- Monitor workloads to avoid overloading team members.

8.4 Monitoring & Reporting

Progress and resource use were tracked using the following activities. Task status uses five categories: Not Started, In Progress, Needs Review, On Hold, and Complete.

Monitoring Activity	Frequency	Responsible
Track task progress	Weekly	Mazen
Prepare project progress report	Weekly	Mazen
Monitor software development	Weekly	Mohamed
Track procurement and expenses	Weekly	Rahma
Check hardware availability	Weekly	Rana

Monitoring Activity	Frequency	Responsible
Review cybersecurity	During development and testing	Gehad
Monitor network and IoT connectivity	Weekly	Asmaa

9. Budget

IHOME is offered in three packages matched to house size and level of automation.

Package	House Size	Price Range	Included Features
Essential	100–150 m²	$5,000 – $7,000	Smart lighting, climate control, and voice commands Basic security system, safety sensors, and mobile app control
Plus	150–220 m²	$7,000 – $11,000	Advanced automation for lighting, curtains, and temperature Smart cameras, child & pet monitoring, and remote control
Premium	220–300 m²	$11,000 – $15,000	Full home automation with AI-based routines and smart entertainment Advanced security, multi-room monitoring, and intelligent energy management

10. Risk Management

Four main risks were identified at the start of the project, each with an assigned owner and a mitigation strategy.

Risk	Description	Impact	Probability	Owner	Mitigation Strategy
Budget Overrun	Project costs may exceed the planned budget due to unexpected expenses or increasing equipment prices.	**Medium**	**Medium**	Project Management	Set a fixed budget, compare multiple suppliers, and monitor expenses regularly.
Technical Issues	Hardware: smart devices might break, sensors could fail to detect movement, or wiring might get damaged. Software: the mobile app might crash, the AI voice assistant could misunderstand commands, or the integration might fail.	**High**	**Low**	Software & Hardware Manager	Conduct regular system testing, use compatible devices and standardized protocols, hire qualified technicians, and prepare backup solutions for critical components.

Risk	Description	Impact	Probability	Owner	Mitigation Strategy
Security Threats	Hackers might try to break into the network to open smart doors or watch the cameras.	**High**	**Low**	Security Engineer	Implement strong authentication, data encryption, network segmentation, regular security updates, and continuous security monitoring.
Privacy Concern	Unauthorized people or apps might access the family's private data, smart cameras, or daily routines.	**Low**	**Medium**	Network Engineer	Apply strict access controls, obtain user consent before collecting data, minimize stored personal information, and use encrypted data storage.

11. Quality Assurance

11.1 Quality Objectives

The Quality Assurance Plan aims to ensure that IHOME meets the required standards of performance, reliability, security, and usability. The project aims to:

- Ensure reliable operation of all smart home devices.
- Provide accurate control of lighting, security, and climate systems.
- Ensure smooth operation of the mobile application and voice control.
- Protect user information and prevent unauthorized access.
- Provide a simple and user-friendly experience.
- Ensure successful integration between all smart home components.

11.2 Quality Criteria

Deliverable	Quality Criteria
Smart Lighting	Lights respond correctly to commands and schedules.
Smart Security	Cameras, sensors, and security alerts function correctly.
Climate Control	Temperature settings are accurate and responsive.
Mobile Application	The application is user-friendly and controls devices correctly.
Voice Control	Voice commands are recognized and executed correctly.
System Integration	All devices communicate and operate together successfully.
Documentation	User manuals and technical documents are complete and clear.

11.3 Quality Methods

Method	Purpose
Functional Testing	Verify that every feature operates according to requirements.
Integration Testing	Ensure that all smart devices communicate correctly.

Method	Purpose
Security Testing	Identify vulnerabilities and prevent unauthorized access.
Usability Testing	Evaluate whether the system is easy to understand and operate.
Performance Testing	Measure system response time and reliability.
User Acceptance Testing	Allow stakeholders to test the system and confirm that it meets their expectations.

11.4 Quality Roles

Team Member	Role	Quality Responsibility
Mazen	Project Manager	Monitor overall quality and ensure project requirements are achieved.
Mohamed	Software & Testing	Test the mobile application and resolve software errors.
Rahma	Business Development	Collect stakeholder feedback and verify that expectations are met.
Rana	Hardware & System Architecture	Ensure hardware components are installed and function correctly.
Gehad	Security Engineer	Perform security checks and protect user data.
Asmaa	Network & IoT	Test network connectivity and communication between devices.

11.5 Quality Monitoring

- Reviewing completed tasks at the end of each sprint.
- Tracking the number of errors and technical issues.
- Checking deliverables against the established quality criteria.
- Conducting regular testing and progress reviews.
- Collecting stakeholder feedback.
- Recording unresolved defects and monitoring their status.

11.6 Corrective Actions

Whenever a quality issue is identified, the team follows these steps:

1. Identify and document the problem.
2. Analyze the root cause of the issue.
3. Assign the issue to the responsible team member.
4. Implement the necessary corrective action.
5. Retest the affected feature to verify the solution.
6. Confirm that the issue has been resolved before approving the deliverable.

12. Communication Plan

All progress reports are directed to the Project Manager, using the following channels and frequencies.

Deliverable	Team	Means	Frequency	Priority	Purpose
Lighting System	Lighting System Team	Team Meeting	Weekly	**High**	Report progress and any technical issues.
Voice Control	Voice Control Team	Email	Weekly	**High**	Demonstrate voice commands and functionality.
Security System	Security Team	Team Meeting	Weekly	**High**	Report security issues and system performance.
Climate Control	Climate Control Team	Email	Weekly	**Medium**	Ensure proper temperature monitoring and control.
Mobile Application	Mobile App Team	Team Meeting	Bi-weekly	**High**	Share development updates and app integration progress.
System Installation	Installation Team	Team Meeting	During installation phase	**High**	Ensure all devices are properly installed and connected.
Testing & Final Documentation	QA & Documentation Team	Email & Team Meeting	End of project	**High**	Submit final testing results and project documentation.

13. Testing & Verification

Final testing takes place in the last 14 days of the project, ending on 17 January 2027, once installation and the cybersecurity review are complete. Each testing method from the Quality Assurance Plan has a clear scope and owner, and any defect found follows the corrective-action steps in Section 11.6.

Test Type	Scope	Owner
Functional Testing	Every feature operates according to requirements across lighting, security, climate, voice control, and the mobile app.	Mohamed, Rana
Integration Testing	All smart devices communicate and operate together through the network and the mobile app.	Asmaa, Mohamed
Security Testing	Vulnerabilities, access control, and data protection are checked to prevent unauthorized access.	Gehad
Usability Testing	The app and voice control are easy to understand and operate.	Mohamed
Performance Testing	System response time (target: under 2 seconds for standard control commands) and reliability.	Mohamed
User Acceptance Testing	Stakeholders test the system and confirm it meets their expectations; feedback is collected for the final handover.	Rahma, Mazen

14. Conclusion

IHOME brings intelligent, secure, and energy-efficient living into the home by integrating lighting, climate control, security, voice control, and a centralized mobile application into the home's infrastructure during construction. The project is organized into seven deliverables, supported by a 10-week schedule, defined responsibilities for each of the six team members, a risk management plan, and a quality assurance process covering functional, integration, security, usability, and performance testing.

Final delivery and stakeholder handover are planned for 17 January 2027, with the system measured against the seven project objectives set out in Section 3.

14.1 Recommendations & Future Work

- Use the expandable infrastructure to add new smart devices (at least 5 are supported without major structural modifications).
- Extend automation to additional home functions such as curtains and appliances.
- Keep security updates and continuous monitoring in place after handover to protect user data and privacy.
- Offer the Essential, Plus, and Premium packages across different house sizes, with the Premium package adding AI-based routines and intelligent energy management.